1.

Formula used: Market Capitalization = Stock Price * Shares Outstanding

a) Calculation for year-end 2024:
   - Stock Price for year-end 2024: $20.05
   - Shares Outstanding: 4,280 million
   - Market Capital: $20.05 * 4,280,000,000 = $85,814,000,000
   - Total 2024 market capital measures about $85.81 Billion (approx.)

b) Calculation for year-end 2023:
   - Stock Price for year-end 2023: $50.25
   - Shares Outstanding: 4,212 million
   - Market Capital: $50.25 * 4,212,000,000 = $211,653,000,000
   - Total 2023 market capital measures about $211.65 Billion

Metric	Year-end 2024	Year-end 2023
Stock Price	$20.05	$50.25
Shares Outstanding	4,280 million	4,212 million
Market Capitalization	$85.81 billion	$211.65 billion
Category identified	Mega Cap (>$100B)	Large Cap ($10B to $100B)

- Determine Market Cap: For the fiscal year ending December 28, 2024, the market capitalization of Intel dropped to around $85.81 billion. This happened after the company posted a substantial net loss of $18.76 billion.
- Match to Chart: Based on your "Categories of Companies According to Their Market Cap" table: Mega Cap is above $100 billion. Large Cap is between $10 billion and approximately $100 billion.
- Conclusion: As $85.81 billion is categorized in the $10B - $100B category, Intel shifted from "Mega Cap" in 2023 to "Large Cap" by the end of 2024.

2. Calculation of the trailing 12 months for Intel.

Price/Equity Ratio = Stock price per share/ Earnings per share

Calculation = $20.05/ (-$4.38) = - 4.58

Result and Analysis:

- The trailing 12 months P/E ratio for Intel as of December 31, 2024, is -4.58.
- Important Note on Negative P/E Ratios: In financial markets, a negative P/E ratio occurs when a company has negative earnings (a net loss) over the specified period. Because Intel reported a significant net loss of $18.76 billion for 2024, the P/E ratio is technically negative.
- In most financial databases and stock screeners, a company with negative earnings is often listed as "N/A" (Not Applicable) or "NE" (No Earnings), as a negative ratio does not provide a meaningful valuation multiple for traditional comparison. This negative figure reflects the company's fiscal challenges during 2024, which also led to its change from a "Mega Cap" to a "Large Cap" company.

Intel's 2024 financial data, Intel is categorized as a **Large Blend** stock:

Valuation Analysis

Using the **20.05** closing price from December 31, 2024, and the **-4.38** diluted EPS, Intel's relative stock valuation compared to the overall market is as follows:

- Negative Valuation Multiple: The trailing P/E multiple for Intel in 2024 is -4.58. This is compared to the overall stock market (which generally keeps a positive P/E ratio, such as the average for the S&P 500). Intel is currently in a "loss-making" phase of valuation.
- Sector Comparison: Although the overall stock market in 2024 experienced growth, Intel's stock price fell dramatically because of a net loss of $18.76 billion. Its stock price is currently reflective of a firm in a "turnaround" or restructuring phase, rather than a growth or value phase.
- Price-to-Sales (P/S) Analysis: Within the "Blend" group, which takes into account both growth and value stocks, the P/S ratio of 1.6 for Intel is below the Information Technology sector average. This indicates that although Intel is currently losing money, its revenue growth is still being assessed by the market at a "deep value" discount compared to similar firms such as NVIDIA or AMD.

Category	Classification	Rationale
Size	Large Cap	Market cap of **$85.81B** is between $10B and $100B.
Style	Blend	Intel exhibits characteristics of both a **Value** stock (low P/S ratio) and a **Growth** stock (significant R&D investment for a turnaround).

3.

Free Cash Flow (FCF) is calculated as: Operating Cash Flow - Capital Expenditures (CapEx).

- **Cash from Operations (2024): $8.29 Billion**
- **Capital Expenditures (2024): $23.94 Billion**
- **2024 Free Cash Flow:** $8.29 - 23.94 = -$15.65 billion

To find value, we must assume Intel returns to profitability as their "18A" manufacturing process matures and they reduce CapEx.

- **Year 1 (2025):** Still negative but improving as cost-cuts take hold $-5B.
- **Year 2 (2026):** Breaking even as new foundry customers sign on ~$0B.
- **Year 3-5 (2027-2029):** Returning to historical norms of roughly **$12B - $15B** in positive FCF.
- **Assumed Terminal Growth Rate: 2.5%** (This is a standard judgement call, reflecting long-term inflation and steady industry growth).

Enterprise Value Calculation:

Present Value (PV) for the next 5 years FCF: $20 billion

Present Value of terminal value: $107 billion

Total enterprise value: $127 billion

Equity Value Calculation:

Total Debt: $50.47 billion

Enterprise Value: $127 billion

Total Equity Value: $76.53 billion

Share Value Calculation:

Total diluted shares: 4.28 billion

Share Value: $17.88

Metric	**Value**
Enterprise Value	$127 billion
(-) Total debt	($50.47 billion)
Implied equity value	$76.53 billion
Shares outstanding	4.28 billion
DCF implied share price	$17.88

Conclusion:

My calculated value of $17.88 is lower than the closing price of $20.05 on Dec 31. This indicates that at a price of $20.05, the market was slightly more optimistic about the speed of Intel's recovery than this particular conservative scenario.